COMP 2711
Discrete Math Tools for Computer Science

Mock Midterm Examination 10

Practice material – not an official examination

Time allowed: 120 minutes Total: 100 points

Name: _____	Student ID: _____
Email: _____	Lecture: _____

Instructions

- This closed-book practice exam consists of 10 problems. You may use one A4 single-sided handwritten cheat sheet.
- Answer all problems in the spaces provided. You may also use the backs of pages and the draft pages. Each problem starts on a new page; some answers require only a few lines.
- Give a proof, justification, or counterexample when requested. State any additional notation you introduce.
- Answers may be left in exact form using binomial coefficients, factorials, and $C(n,k)$ or $P(n,k)$. For example, $\binom{5}{3} + \binom{4}{2}$ is acceptable. Avoid nonstandard notation such as nCk or nPk.
- The separate solutions document contains derivations and marking notes. Attempt the exam before consulting it.

Problems	1	2	3	4	5	6	7	8	9	10	Total
Points	10	13	12	16	10	9	7	6	7	10	100
Score											

Problem 1: Propositional logic and proofs [10 points]

(a) **[3 points]** Classify with a logical-law justification: (i) $(p \to q) \vee (q \to p)$; (ii) $(p \leftrightarrow q) \wedge (p \oplus q)$; (iii) $(p \to q) \wedge \neg q$. Each is worth 1 point.

(b) **[2 points]** Is the liar sentence *This sentence is false* a proposition under the classical definition used in the lecture (1 point)? If $p \to q$ is given, does that logically assert q *only if* p (1 point)? Explain briefly.

(c) **[5 points]** Give direct proofs: (i) if u, v are rational and $v \neq 0$, then u/v is rational (3 points); (ii) if an integer n is odd, then $n(n+2)$ is odd (2 points).

Problem 2: Predicate logic [13 points]

(a) **[4 points]** In $\forall x P(x) \vee Q(x)$, use quantifier precedence to insert parentheses, identify free occurrences and decide whether it is a proposition (2 points). Translate *some real is at least every real* and give its truth value with justification (2 points).

(b) **[4 points]** Let A be a proposition with no free x, and first assume the domain is nonempty. Prove $\forall x(P(x) \to A) \equiv ((\exists x P(x)) \to A)$ (2 points). Now let the domain be empty: give a truth value of A showing that $\forall x(A \wedge P(x)) \equiv A \wedge \forall x P(x)$ need not hold (2 points).

(c) **[5 points]** From $\forall x(P(x) \to Q(x))$, $\forall x(R(x) \to \neg Q(x))$, and $\exists x(R(x) \wedge S(x))$, prove $\exists x(S(x) \wedge \neg P(x))$ using inference rules.

Problem 3: Sets and functions [12 points]

Use $\mathbb{N} = \{0, 1, 2, \ldots\}$.

(a) **[2 points]** Within $U = \{0, 1, 2, 3, 4, 5\}$, let $A = \{0, 2, 4\}$ and $B = \{2, 3, 4\}$. Find $A \setminus B$, $U \setminus (A \cup B)$, and $|A \cup B|$, verifying the last value by the two-set identity.

(b) **[3 points]** Construct an explicit bijection $h : \mathbb{N} \to \mathbb{Z}$ by assigning nonnegative integers to even inputs and negative integers to odd inputs. Prove injectivity and surjectivity.

(c) **[7 points]** Write T or F for each statement. No justification is required. Each item is worth 1 point.

(i) $\mathbb{N} = \{0, 1, 2, \ldots\}$ is finite.

(ii) The nonnegative even integers are countably infinite.

(iii) The set of binary sequences with exactly six ones is countably infinite.

(iv) $\mathbb{Q} \cap (0, 1)$ is countably infinite.

(v) Every proper subset of a finite set has smaller cardinality.

(vi) An uncountable set can be listed by the natural numbers with no omissions.

(vii) The set of all finite computer programs is uncountable.

Problem 4: Short counting calculations [16 points]

Give exact answers. Factorials, powers, and binomial coefficients may be left unevaluated. No explanation is required unless requested.

(a) **[2 points]** Evaluate $\sum_{k=0}^{6} \binom{6}{k}^2$ using a binomial identity.

(b) **[2 points]** Eight distinct people sit around a circle with three specified people consecutive in any internal order. Rotations are equivalent; reflections are different. Count the seatings.

(c) **[2 points]** Count strings using exactly two As, two Bs and two Cs, with the two As not adjacent. Equal symbols are indistinguishable.

(d) **[2 points]** Choose an unordered six-person committee from ten distinct people, with A and B not both selected. Count the committees.

(e) **[2 points]** Choose six items from three unlimited types A, B, C, with an even number of type A. Items of a type are identical and order does not matter. Count the choices.

(f) **[2 points]** Find the constant term in $(x - 2x^{-1})^8$.

(g) **[2 points]** Count injective functions $f : \{1, 2, 3, 4\} \to \{1, 2, 3, 4\}$ with $f(2) = 2$ and $f(1) \neq 1$.

(h) **[2 points]** Count right/up unit-step lattice paths from $(0, 0)$ to $(4, 3)$ that do not begin with two right steps.

Problem 5: Combinatorial proof [10 points]

[10 points] For $n \geq 1$, use a complement bijection to prove

$$\sum_{k=0}^{n-1} \binom{2n}{k} = \frac{2^{2n} - \binom{2n}{n}}{2}.$$

Explain why the subsets of size n must be treated separately.

Problem 6: Inclusion-exclusion [9 points]

[9 points] Use inclusion–exclusion to count integers in $\{1, \ldots, 180\}$ divisible by exactly two of $2, 3, 5$. Explain how your expression removes integers divisible by all three.

Problem 7: Pigeonhole principle [7 points]

[**7 points**] Prove that any seventeen distinct bit strings of length eight include three with the same number of ones. Show that seventeen is the smallest number with this guarantee.

Problem 8: Joint masses and event rules [6 points]

An outcome is an ordered bit pair. Its probabilities are $P(00) = 1/2$, $P(01) = 1/4$, $P(10) = 1/8$, $P(11) = 1/8$. Let A mean first bit one and B mean second bit one.

(a) **[1 points]** Find $P(A)$ and $P(B)$.

(b) **[1 points]** Find $P(A \cap B)$.

(c) **[1 points]** Find $P(A \cup B)$.

(d) **[1 points]** Find $P(A^c \cap B^c)$.

(e) **[2 points]** Justify the two-event union rule by a disjoint-set decomposition, and test independence of A, B.

Problem 9: Restricted bit strings [7 points]

Choose uniformly among the sixteen bit strings of length four. Let E mean first bit one, F mean exactly two ones, and G mean that the two ones are consecutive (so $G \subseteq F$).

(a) **[3 points]** Find $P(E \mid F)$ and $P(G \mid F)$ using the restricted sample space.

(b) **[4 points]** Find $P(F \mid E)$ and use it to test independence of E, F. State the positivity requirement for this conditional test.

Problem 10: Bernoulli sums and variance proofs [10 points]

Let $I_1, \ldots, I_n$ be independent Bernoulli indicators with success probability p, where $n \geq 1$ and $0 < p < 1$. Put $S = \sum_i I_i$ and $M = S/n$.

(a) **[2 points]** Compute $E(I_i)$ and $V(I_i)$ from its two possible values.

(b) **[2 points]** Find $E(S)$ and $V(S)$, identifying where independence is used.

(c) **[2 points]** Find $E(M)$ and $V(M)$.

(d) **[4 points]** Derive $V(X) = E(X^2) - E(X)^2$ for a finite random variable. Then prove $V(X + Y) = V(X) + V(Y)$ when X, Y are independent finite variables.

Draft Paper

Mock examination working space.

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